

Experimental comparison and deformation assessment

Rigid-body linear and independent angular impulses

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Experimental comparison and physical scope

We do not yet have a model validated across the real-life datasets. This report covers every presently recovered comparison record: 24 glass collisions, eight oblique ball/surface summaries, 75 rock impacts, and 17 rolling measurements. These are public experimental records, not invented target states. They are not 124 independently characterized full-state events: some coefficients summarize overlapping characterization data. Numerical verification and experimental agreement are separate questions.

Documented parameters remain fixed. Missing parameters are estimated only in small, explicit calibration models. Same-outcome fitting is not prediction. Every unfavorable result is retained. The full engine still lacks validated rolling/torsional material closure and has unresolved rapid irregular-contact cases. No 2x performance gate is applied; the user removed it.

Comparison	Records	Outcome	Qualification
Glass fixed-input comparator	24	0.0367 / 0.0178 m/s RMSE	No independent spin
Ball endpoint/moment test	8	4 passive rolling nominal fits	Singleton calibration
Shared moment, held-out surface	8	RMSE 1.048 to 1.243 rad/m	Worse; rejected
Rock sphere-proxy comparator	50 + 25	Spin RMSE 12.88 rad/s	Actual shape absent
Tennis rolling, held-out speed	9 + 8	Constant worse than source curves	Quasirolling only
Isolated rolling mechanics	200	All pass at three length scales	Synthetic controls

The strongest result is a consistent impulse/momentum framework including an independent angular impulse. That framework alone does not specify authentic contact forces. Energy-passive estimates can still predict poorly; this is demonstrated here. No comparison in this report establishes full 3D actual-shape validation with measured signed angular-velocity vectors and independently measured inertia tensors.

Mechanics used and physical checks

The notation retains the user's combined variables and inertia symbol:

$$\mathbf{V} = \begin{bmatrix} \mathbf{v} \\ \ell \boldsymbol{\omega} \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} \mathbf{p} \\ \mathbf{L}/\ell \end{bmatrix}, \quad \Delta \mathbf{P}_c = \begin{bmatrix} \delta \mathbf{p} \\ \delta \mathbf{L}/\ell \end{bmatrix}.$$

The full angular update is

$$\Delta \mathbf{L}_k = \mathbf{r}_k \wedge \delta \mathbf{p} + \delta \mathbf{L}.$$

The independent couple is not the lever-arm moment counted twice. In the ball calculations the input has no spin and the plane fixes the onset axis of generated spin. This is a declared planar onset branch, not a recovered general mixed-spin law. All scalar directional components remain in the recorded output.

Keep $\hat{\mathbf{t}}$ along full relative contact velocity. When it contains normal approach, the source's normal/tangential friction must be transformed through the direction map. With $c = \hat{\mathbf{n}}^\top \hat{\mathbf{t}}$,

$$\hat{\mathbf{n}}^\top \delta \mathbf{p} = \delta p_n + c \delta p_t, \quad \|Q \delta \mathbf{p}\| = \sqrt{1 - c^2} |\delta p_t|, \quad Q = \mathbb{1} - \hat{\mathbf{n}} \hat{\mathbf{n}}^\top.$$

The report code checks the full impulse reconstructed from this nonorthogonal direction map; it does not silently redefine $\hat{\mathbf{t}}$ as another direction.

For pure rolling the physical moment length is $a_r = R$:

$$|\delta L_s| \leq \mu_r R \hat{\mathbf{n}}^\top \delta \mathbf{p}, \quad |\delta L_s/\ell| \leq \mu_r R/\ell \hat{\mathbf{n}}^\top \delta \mathbf{p}.$$

With $\mathbb{I} = \alpha m R^2$, no-slip motion slows at $\mu_r g/(1 + \alpha)$ and needs static capacity $\mu_s \geq \mu_r/(1 + \alpha)$. Sliding dissipation is zero on that branch. The torque supplies dissipation while static friction supplies the reaction. The coordinate length ℓ is not a physical deformation/contact length.

The isolated rolling checks cover 100 planar and 100 arbitrarily rotated spatial cases, three representation lengths each, angular arrest, a zero-torque control, independent Newton/Euler balances, and rejection when static capacity is insufficient. Maximum relative motion, energy and scale errors are approximately 1.1×10^{-15} . These do not validate the production collision solver.

Separate material parameter table

The catalog has 11 published normal/tangential/sliding triples. It does not provide 11 complete profiles for the expanded static/dynamic/rolling model. A missing static coefficient is not set equal to a sliding coefficient. A missing rolling coefficient is not called zero measured friction. The legacy catalog and native results remain historical comparator evidence.

Profile	e.n	e.t	Sliding mu	mu.s	mu.r
delrin-smallparts-binary	0.970	0.540	0.140	missing	missing
chrome-steel-binary	1.000	0.540	0.120	missing	missing
glass-soda-binary	0.970	0.440	0.092	missing	missing
glass-fresh-binary	0.922	0.370	0.048	missing	missing
glass-spent-binary	0.972	0.250	0.177	missing	missing
acetate-binary	0.870	0.430	0.250	missing	missing
polyurethane-binary	0.600	0.300	0.800	missing	missing
glass-soda-aluminum	0.831	0.310	0.125	missing	missing
acetate-aluminum	0.891	0.390	0.208	missing	missing
steel-zerodur-air	0.970	0.340	0.110	missing	missing
glass-zerodur-air	0.970	0.390	0.106	missing	missing

The glass test uses its unchanged documented values. None of the listed missing angular/static inputs is identified by those 24 records. Public coefficients apply to their specified material pair, specimen size, condition and measurement convention. Rubber compounds, felt age, surface roughness and support compliance cannot be erased by labeling a parameter “rubber friction.”

Dimensionless rolling coefficients and dimensional moment lengths must be distinguished. Fuchs et al.’s micro-sphere measurements support this normalization, but their measured rolling resistance combines two contacts and adhesion. It is not a macroscopic rubber or single glass/silicon collision profile. No such transfer is used here.

Glass collisions: fixed parameters versus measured output

All 24 glass worksheet records are recalculated using the nominal sphere mass/radius, homogeneous inertia and the documented coefficients. The baseline has zero free couple. Fresh analytical endpoints agree with the preserved native runs to 10^{-7} m/s. All predictions pass total-energy accounting. The measured outgoing normal velocity and author-reconstructed COM tangent velocity are compared row by row in the appendix.

Output	RMSE	Interpretation
Normal velocity	0.036674 m/s	Experimental residual
COM tangent velocity	0.017805 m/s	Experimental residual
Contact tangent velocity	0.062317 m/s	Spin reconstructed by source

The worksheet reconstructs contact spin using angular momentum. Consequently its contact tangent output is not an independently measured spin vector. The coefficient chart may share its characterization trials with the worksheet. This is reproduction using published inputs, not independent cross-condition material validation.

The nominal diameter and homogeneous tensor are modeling approximations. We do not fit an independent couple to author-reconstructed spin and then claim its agreement as proof that the real angular impulse was measured.

Ball spin: fixed restitution and a held-out moment test

For each row the measured normal and tangential restitution are held fixed. The homogeneous-sphere inertia factor is assumed, not measured. The spin factor $S = \omega^+ / v^-$ is the tested output. At zero incoming spin and incidence angle θ to the normal, an independent impulse with signed moment length d gives

$$S = \frac{(1 + e_t) \sin \theta - d/R(1 + e_n) \cos \theta}{(1 + \alpha)R}.$$

First test $d = 0$. Then fit one shared d/R per ball using three surfaces and predict the fourth; repeat for all surfaces. A signed pressure moment is distinct from negative rolling friction. The constant cross-surface hypothesis is not accepted merely because each prediction passes a total-energy check.

Ball / surface	e.n	e.t	Measured S	d=0	Held out S
golf / granite	0.87	0.03	14.8	14.529	15.290
golf / rubber	0.59	0.51	20.5	21.300	22.158
golf / superball.disk	0.81	0.55	23.6	21.864	22.091
golf / tennis_strings	0.91	-0.01	15.0	13.965	14.453
superball / granite	0.78	0.49	14.9	15.510	15.918
superball / rubber	0.78	0.41	14.5	14.677	14.948
superball / superball.disk	0.78	0.57	18.2	16.343	15.967
superball / tennis_strings	0.91	-0.10	9.0	9.368	9.751

Spin RMSE worsens from 1.048 to 1.243 rad/m; worst error worsens from 1.857 to 2.233 rad/m. This shared correction is rejected.

This is conditional endpoint validation: source restitution comes from the same experiment, while each held-out spin is unused in that fold's new fit. It is not independent validation of the supplied restitution/friction material values. In particular the source's e_t itself contains the measured outgoing spin. Withholding that spin from the new fit is therefore not a blind outgoing-state test.

What the missing angular parameters would have to be

Estimating a separate moment from each measured spin explains that row algebraically, but supplies no held-out test. A purely dissipative rolling model requires $\mu_r \geq 0$. At nominal inputs four rows require an assisting couple; fitting a negative friction coefficient is therefore rejected. This is a necessary endpoint sign test, not a verified rolling-force history. A deforming elastic patch can return energy via an assisting moment without violating total passivity. A positive total-energy check alone does not establish its traction distribution or true contact size.

Ball / surface	Needed μ_r	Offset d (mm)	Passive rolling at nominal
golf / granite	-0.00479	-0.102	no
golf / rubber	0.01664	0.356	yes
golf / superball_disk	-0.03170	-0.678	no
golf / tennis_strings	-0.01791	-0.383	no
superball / granite	0.01535	0.445	yes
superball / rubber	0.00446	0.129	yes
superball / superball_disk	-0.04674	-1.356	no
superball / tennis_strings	0.00864	0.251	yes

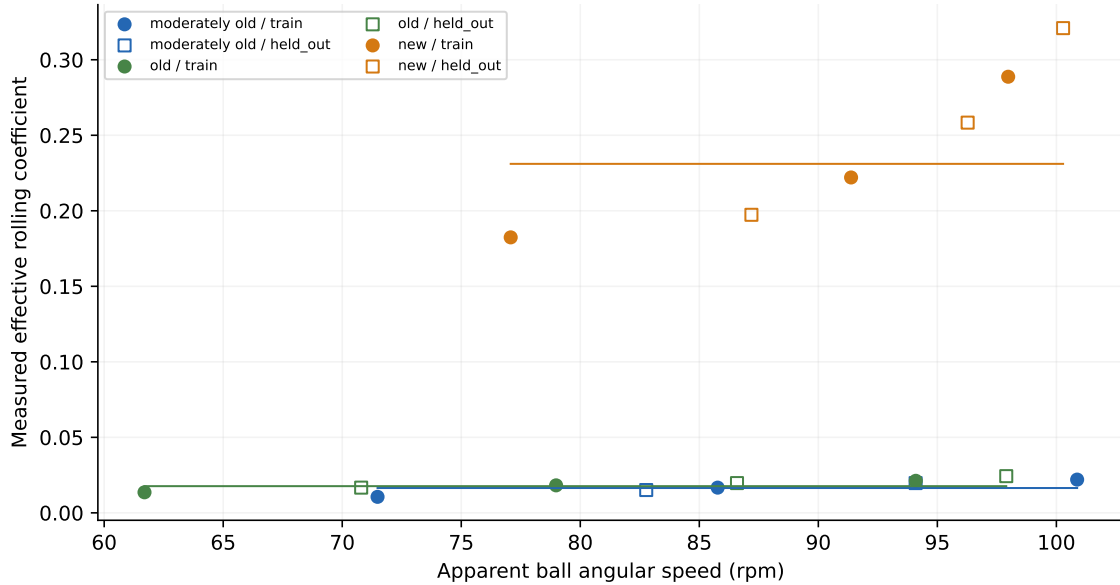
The offset column is a signed inverse-mechanics diagnostic, not measured geometry. Its magnitude lies within the sphere radius and the reconstructed total outgoing energy is below incident energy for these rows. The tighter actual patch-radius bound is unknown. Inertia and force-history errors can mimic the same moment, so these endpoints do not uniquely identify rolling resistance.

Published incidence, restitution and spin error limits are propagated into the JSON envelope. They are maximum-error limits, not statistical confidence intervals. An incompatibility at a nominal point must not be interpreted as a confident material rejection when that envelope includes zero. No independent μ_s or μ_d is invented from the net impulse ratio; that ratio is only a necessary integrated capacity diagnostic for a given force history.

Tennis rolling: estimate missing coefficients and test speed dependence

Experimental circle centers are extracted from the primary PDF's three plots, not from its regression curves. Duplicate PDF paths are removed. Nine alternating speed points calibrate one positive constant per specimen; eight other points test it. The author's published curves remain fixed and are a separate comparator.

Specimen	Estimated μ_r	Held-out RMSE	Fixed curve RMSE
moderately_old	0.01644	0.002495	0.000734
old	0.01767	0.004042	0.000802
new	0.23107	0.057620	0.022930



Circles are calibration points; open squares are held-out points. Horizontal lines are fitted constants. Each constant improves over an absent rolling moment, but each loses to the fixed speed-dependent curve. A universal constant is especially inadequate for the new specimen. This motivates rate-dependent deformation loss, not an arbitrary per-impact correction.

The source curves were fitted by their authors to these published measurements, including the points withheld from our constant fit. Their smaller residual is source reproduction, not independent evidence of predictive superiority. Only the constant-fit residual has the declared local calibration/test separation.

The apparatus measures quasirolling with slight skid. The estimates are restricted to the reported apparent-speed ranges and specimens. Ball radii/inertia and complete matched impact parameters are not supplied. No outgoing bounce state is validated by these rolling points. One-PDF-point digitization sensitivities and leave-one-out estimate ranges are retained in the results. Negative source intercepts are not extrapolated into negative material resistance.

Rocks and non-spherical collision scope

The 75 limestone/concrete records are genuine experimental magnitudes. The earlier three-parameter sphere surrogate is recomputed unchanged, with its original 50-record calibration and 25-record release-height holdout. These project estimates are not independently documented material values. The user now permits small fits, but this does not make a fitted shape proxy an authentic rock model.

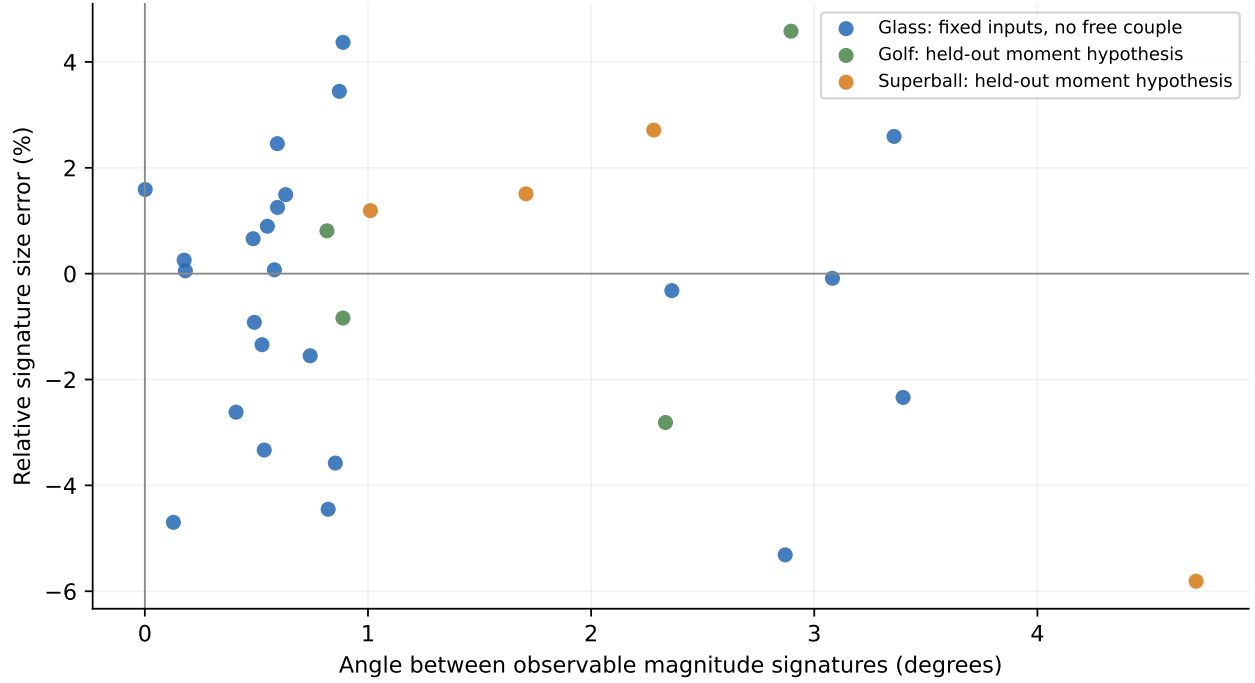
Held-out quantity	RMSE
Normal velocity	1.0044 m/s
Tangent velocity	1.3047 m/s
Angular speed	12.8765 rad/s

Exact collision mesh, central tensor, attitude and signed angular vectors are absent. Zero incoming spin and coplanar signed reconstruction are assumptions. A conditional inverse moment is recorded under those same assumptions, but it cannot identify a true rolling coefficient. All 75 row values remain in the appendix/CSV; failures are not removed.

The synthetic rotated cuboid, irregular 2D/3D controls and large rapid worlds are numerical tests. They cannot replace non-spherical experimental validation. The 13-case rapid/irregular baseline remains unqualified. Production still rejects nonzero rolling/twisting inputs rather than silently accepting an absent mechanism. The 200 new rolling checks concern a separate sustained-contact branch.

One deviation diagram, with the observations stated

Each dot compares a predicted outgoing magnitude signature with a measured one. Horizontal position is the angle between signatures; vertical position is the relative difference in signature norm. A perfect match is at $(0, 0)$. This angle is not a physical heading error.



Glass uses the two observed translational components. Balls use normal velocity, tangent velocity and $R|\omega|$; tangent velocity is reconstructed from source tangential restitution and measured spin. Ball dots use the held-out signed-moment hypothesis, not same-point exact fits. The signature channels differ, so no pooled accuracy percentage is computed. Rock dots are excluded from this fresh diagram because its actual geometry/directional model is not validated.

The standalone HTML plot permits hovering to identify a case. Machine-readable coordinates, definitions and complete records accompany the PDF. Historical bar figures are not used as the new model's validation.

Deformation and vibration: the supported next model

Use a finite-duration deformable patch with a small set of internal modes. An instantaneous rigid impulse cannot describe the measured stretch, force reversal, pressure migration and residual vibration. Begin with normal indentation, tangential shear and a rocking/asymmetric-pressure mode, with positive mass/stiffness and nonnegative damping. A distributed patch supplies the independent angular impulse:

$$\delta \mathbf{p} = \int \sum_k \mathbf{f}_k dt, \quad \delta \mathbf{L} = \int \sum_k \mathbf{x}_k - \mathbf{x}_c \wedge \mathbf{f}_k dt.$$

The body's angular update still contains both $\mathbf{r} \wedge \delta \mathbf{p}$ and this independent $\delta \mathbf{L}$. Internal vibration velocities contribute to each patch point's actual slip. Friction must use that slip, not only an undeformed rigid contact velocity. The general component map must be checked against those resultants; silently projecting away an unrepresented moment is unacceptable.

For modal coordinates $\mathbf{q}$, a compact material model is

$$M_q \ddot{\mathbf{q}} + C_q \dot{\mathbf{q}} + K_q \mathbf{q} = H^T \mathbf{f}, \quad \mathbf{u}_c = B\mathbf{V} + H\dot{\mathbf{q}}.$$

Use the gradient of one elastic potential for contact force and moment. Its normal derivative must be retained if shear stiffness depends on compression. Track rigid kinetic energy, modal kinetic energy, stored strain and dissipation; do not delete stored energy when the contact opens. Integrating then yields effective restitution; do not impose a second endpoint impulse on top of a resolved compliant collision.

Hertz/Mindlin contact and local stick/slip provide an established starting point for small elastic spheres. Large rubber deformation needs a hyperelastic/viscoelastic shell or solid description, then modal reduction. Pressure history can produce rolling resistance; adding a fitted rolling torque to an already resolved loss mechanism can double-count dissipation. Retain $\mu_r a_r$ only for unresolved loss.

How to select and validate the deformation model

The Cross 2014 author manuscript contains measured force/spin histories and separate vibration tests for a hollow ball and a superball. It supports this mechanism, but its force plate and granite measurements use different counterfaces. Their friction values must not be mixed. Attached-ball vibration modes also use different boundary conditions from a free bounce; their frequencies constrain, rather than uniquely determine, the dynamic contact stiffness. [5]

2014 specimen / pair	Mass	Diameter	Source sliding μ
Hollow ball / granite	44.2 g	59.4 mm	not reliably measured
Solid superball / granite	46.2 g	46.0 mm	0.45 +/- 0.08
Hollow ball / G10 force plate	same specimen	same specimen	approximately 1.8

The reported hollow-ball attached-mode periods are approximately 11 ms normal and 5 ms tangential; the solid ball has a 4.5 ms normal period and 3.5 ms high-frequency tangential period, plus a slower mode. These are separate from the 2010 58 mm Superball and are not automatically a controlled size-only comparison. Static friction and modal damping still require independent estimates.

Keep geometry, independently measured modal periods and pair friction fixed where available. Identify at most a few missing stiffness/damping or patch parameters from normal force histories and independent vibration decay, then reserve oblique spin and velocity outputs. Fit a mode's damping from its decay, not from every desired outgoing spin. Use one parameter set over speed, incidence and incoming spin.

First qualify normal force/time and deformation, then tangential force sign changes, then the independent moment/angular response. Compare full histories and endpoints, including worst-case errors, static/sliding transitions and energy. An apparent endpoint improvement that breaks force histories, restitution inputs or total passivity is rejected. Test a small patch-resolution ladder and a time-step ladder. If two or three modes cannot reproduce those independent checks, obtain a converged finite-element shell/solid reference and reduce its verified modes.

This is a literature-supported implementation direction, not a demonstrated new data fit. The earlier constant shear/rocking candidate worsened held-out error by 23.7%; the new shared moment estimate also worsens it. Neither is adopted. Merely adding another oscillator and tuning it to every endpoint is not sufficient.

Reproduction: run `calculate.py` into a fresh output directory, run `audit.py` against it, `render.py`, then `build.sh`. The authoritative evidence is `evidence-v3`; `v1` and `v2` are preserved preliminary checkpoints. Source snapshots, SHA-256 values, all records and the independent audit accompany the report. No generic material lookup can yet promise authentic full-motion predictions across the requested cases.

Keep rigid bodies: a small contact-memory matrix

The latest user request favors a contact model that encapsulates local deformation without deformable meshes for all bodies. Keep the existing rigid-body inertia and contact mobility. Add elastic history $\boldsymbol{\eta}$, a small positive stiffness matrix K , and nonnegative damping matrix C at an active contact. Angular history uses ℓ times angular displacement, dual to the angular impulse divided by ℓ .

For a frozen elastic sticking branch and step h , the new research core uses

$$\begin{aligned} Z &= \tfrac{1}{2}h^2K + hC, \\ [W_d + 2Z^{-1}]\boldsymbol{\lambda} &= -2\mathbf{z}^- - 2Z^{-1}hK\boldsymbol{\eta}^-, \\ \mathbf{z}^+ &= \mathbf{z}^- + W_d\boldsymbol{\lambda}, \\ \boldsymbol{\eta}^+ &= \boldsymbol{\eta}^- + h\mathbf{z}_m, \quad \mathbf{z}_m = \tfrac{1}{2}\mathbf{z}^- + \tfrac{1}{2}\mathbf{z}^+. \end{aligned}$$

Only a small contact matrix is factored. For constant geometry, step and material it is cached. Body scatter/gather remains the same impulse update. Linear and independent angular channels may couple through K , while the resulting impulse components remain $\delta p_n, \delta p_t, \delta L_s/\ell, \delta L_n/\ell$. The energy balance of this branch is

$$\Delta E + \Delta U + h\mathbf{z}_m^\top C\mathbf{z}_m = 0, \quad U = \tfrac{1}{2}\boldsymbol{\eta}^\top K\boldsymbol{\eta}.$$

This core is implemented in `contact_memory.py` and independently checked with 100 planar and 100 spatial body/contact systems, including 20 dependent-direction cases. It preserves energy/work and representation-length invariance and shows second-order convergence to an exact oscillator solution. The physical energy error is below 1.5×10^{-15} relative in those controls.

Opening, Coulomb yielding, changing directions and nonlinear indentation are not yet implemented in this new core. It is a verified elastic branch, not a production collision law or a demonstrated experimental improvement. Contact history cannot simply be discarded at separation; retained internal vibration or explicit loss must account for its energy. This is the cheap mechanism to extend, rather than a per-body deformable mesh. The current Python four-channel cached step measures 45.4 microseconds median locally. This includes validation and Python overhead; it is not a native end-to-end engine benchmark or an all-case performance guarantee.

Rock friction: pressure and plasticity rather than raw impulse powers

Indentation, asperity crushing and interlocking can add apparent resistance beyond a sliding-test coefficient. That possibility does not establish which mechanism dominates these rock records. Keep documented μ_s, μ_d as the underlying pair coefficients. Model an additional passive deformation/ploughing contribution only when it is independently constrained. Do not silently call net impulse divided by normal impulse a measured dynamic coefficient.

A physically scaled candidate variable is maximum indentation relative to local effective curvature, $\chi = \delta_{\max}/R$. For an elastic Hertz reference with effective modulus E^* , coefficient $k_H = 4E^*\sqrt{R}/3$ and incident normal energy E_n , the estimated compression is

$$U_n = \frac{2}{5}k_H\delta^{5/2}, \quad \delta_{\max} = \left[\frac{5E_n}{2k_H}\right]^{2/5}.$$

This is a local proxy valid for small elastic contact, not a solution for crushing angular rock facets. A yield/history variable is needed once that approximation breaks down. Thornton–Ning provides an established elastic/plastic normal-contact route; LAMMPS granular contact demonstrates practical normal, shear-history, rolling and torsion laws with rigid particles. [10,11]

An $a\chi + b\chi^2$ correction can be tested as a bounded approximation to extra resistance or a moment-length response over a declared range. It is a hypothesis, not a proven material law. Fit at most one or two missing coefficients to a training condition, hold material friction fixed, and reject it if held-out velocity/spin, worst-case error or passivity worsens. Check identifiability against contact size, yield and initial-spin uncertainty.

Unnormalized impulse-polynomial coefficients have units and can change meaning with mass, size and the interval over which impulse is accumulated. For a resting contact that interval changes with timestep. If an impulse variable is used, it must refer to the complete impact and an independently specified physical impulse scale, not the arbitrary coordinate length ℓ . Pressure, contact work or χ offer a clearer route to transferring coefficients across sizes. No pressure-dependent rock law has been adopted or fitted in the reported comparisons.

Primary sources and current limitations

- [1] [Cornell impact coefficients](#)
- [2] [Original glass worksheet](#)
- [3] [Cross 2010: ball/surface experiments](#)
- [4] [Wang et al. 2018: rock impacts](#)
- [5] [Cross 2014: oblique rubber bounce, author manuscript](#)
- [6] [Singh et al. 2008: rolling experiment](#)
- [7] [Fuchs et al. 2014: rolling/sliding/torsion](#)
- [8] [Maw, Barber and Fawcett 1976: elastic oblique contact](#)
- [9] [NASA 2005: measured rolling and failed bounce extension](#)
- [10] [LAMMPS granular contact: history, rolling and torsion](#)
- [11] [Thornton–Ning 1998: elastic/plastic normal contact](#)

Full measured signed 3D states and central tensors are not available for the recovered rock fixtures. Dynamic/static/rolling profiles remain incomplete for most pairs. Published coefficient charts, worksheet reconstructions and singleton summary rows cannot independently identify every angular/material channel. Those limits constrain what the present numbers establish; they are not hidden defaults.

Appendix: all 24 glass collision comparisons

Excel row	Normal obs.	Normal pred.	Tangent obs.	Tangent pred.
2	0.92521	0.97668	-0.48443	-0.44054
13	0.86685	0.80824	-0.63661	-0.67081
24	0.74446	0.73741	-0.74531	-0.75337
35	0.85223	0.89104	-0.60724	-0.61467
46	0.56947	0.61812	-0.93034	-0.89759
57	0.69712	0.69487	-0.78925	-0.77232
68	1.02769	0.95319	-0.39398	-0.42130
79	0.91821	0.92967	-0.63660	-0.63296
90	0.99824	0.98450	-0.53381	-0.53736
101	0.96391	0.93751	-0.54760	-0.58497
112	1.07357	1.02150	-0.30496	-0.30607
123	1.14587	1.19960	-0.23465	-0.22634
134	1.12873	1.14806	-0.22810	-0.21887
145	1.16067	1.17914	-0.22808	-0.23167
156	1.17884	1.19401	-0.04548	-0.03366
167	1.21576	1.15873	-0.04107	-0.03655
178	1.13198	1.16148	-0.16552	-0.15756
189	1.16999	1.12545	0.17683	0.18729
200	1.18091	1.19323	0.18636	0.17660
211	1.12379	1.08438	0.21220	0.21526
222	1.21562	1.21592	0.11036	0.11427
233	1.25086	1.23072	0.04908	0.06424
244	1.10217	1.07165	0.23806	0.23948
255	1.16525	1.16854	0.10528	0.10196

All values are m/s. Tangent output is the source-reconstructed COM quantity. No independently measured spin is added.

Appendix: rock records 1–25

Row	Split	v_n obs/pred	v_t obs/pred	omega obs/pred
18	train	2.570 / 2.645	2.898 / 2.807	48.40 / 42.15
19	train	1.714 / 2.836	2.824 / 2.610	43.20 / 39.20
20	train	1.929 / 2.612	2.902 / 2.718	41.50 / 40.83
21	train	2.826 / 2.684	2.655 / 2.673	49.20 / 40.14
22	train	2.536 / 3.231	3.342 / 3.154	50.60 / 47.36
23	train	2.683 / 3.113	4.008 / 3.376	55.40 / 50.70
24	train	2.184 / 3.293	4.168 / 3.152	44.80 / 47.34
25	test	3.039 / 3.821	3.947 / 3.465	51.30 / 52.04
26	test	3.177 / 3.676	4.074 / 3.784	65.80 / 56.83
27	test	2.718 / 3.895	4.537 / 3.740	59.20 / 56.16
28	train	2.141 / 1.968	3.879 / 3.621	42.60 / 54.37
29	train	2.562 / 2.087	3.852 / 3.944	53.70 / 59.23
30	train	2.040 / 2.033	4.628 / 3.715	39.80 / 55.79
31	train	1.952 / 1.930	3.739 / 3.944	46.40 / 59.22
32	train	2.723 / 2.440	4.468 / 4.545	58.00 / 68.25
33	train	2.406 / 2.360	5.258 / 4.559	53.20 / 68.46
34	train	2.403 / 2.397	4.248 / 4.727	51.60 / 70.99
35	test	2.696 / 2.757	4.682 / 5.122	67.60 / 76.92
36	test	3.722 / 2.809	4.509 / 5.262	58.20 / 79.03
37	test	3.342 / 2.723	5.349 / 5.148	54.10 / 77.31
38	train	2.074 / 1.242	3.755 / 4.450	66.80 / 66.82
39	train	2.230 / 1.414	4.220 / 4.427	59.40 / 66.49
40	train	1.951 / 1.426	4.234 / 4.733	61.70 / 71.07
41	train	2.797 / 1.667	6.323 / 5.680	62.40 / 85.30
42	train	2.713 / 1.708	4.828 / 5.500	81.30 / 82.60

Velocities are m/s; angular speed is rad/s. Prediction is the historical sphere proxy, not actual rock geometry. “test denotes the original release-height holdout.

Appendix: rock records 26–50

Row	Split	v_n obs/pred	v_t obs/pred	omega obs/pred
43	train	3.613 / 1.722	3.849 / 5.314	75.40 / 79.80
44	test	2.016 / 2.161	5.663 / 6.023	64.30 / 90.45
45	test	2.913 / 1.878	5.374 / 6.150	81.50 / 92.36
46	test	1.414 / 1.872	5.394 / 5.917	66.20 / 88.86
47	train	1.582 / 0.676	3.192 / 5.040	75.90 / 75.69
48	train	1.553 / 0.814	3.378 / 4.680	68.20 / 70.29
49	train	1.158 / 0.623	3.367 / 4.839	73.10 / 72.68
50	train	1.641 / 0.705	4.421 / 5.941	91.20 / 86.69
51	train	1.110 / 0.722	7.117 / 6.056	53.50 / 88.68
52	train	1.786 / 0.616	3.551 / 6.359	85.60 / 75.72
53	test	1.663 / 0.641	5.281 / 7.182	97.50 / 78.81
54	test	3.054 / 0.757	3.924 / 7.353	107.20 / 92.99
55	test	3.176 / 0.891	4.014 / 6.905	93.20 / 103.70
57	train	2.119 / 2.645	2.728 / 2.507	20.60 / 18.82
58	train	2.337 / 2.789	3.326 / 2.830	19.70 / 21.25
59	train	2.073 / 2.647	3.030 / 2.642	24.80 / 19.84
60	train	2.289 / 3.233	3.281 / 3.169	29.20 / 23.80
61	train	1.908 / 3.396	4.134 / 3.063	23.00 / 23.00
62	train	1.911 / 3.162	3.510 / 3.268	27.90 / 24.54
63	test	2.833 / 3.627	4.664 / 3.924	30.40 / 29.46
64	test	3.058 / 3.722	3.853 / 3.540	31.60 / 26.58
65	test	2.356 / 3.663	3.725 / 3.362	25.80 / 25.25
66	test	2.869 / 3.854	3.728 / 3.661	26.30 / 27.49
67	train	2.769 / 2.272	3.596 / 3.682	33.80 / 27.65
68	train	2.334 / 2.052	3.156 / 3.508	27.40 / 26.34

Velocities are m/s; angular speed is rad/s. Prediction is the historical sphere proxy, not actual rock geometry.
“test denotes the original release-height holdout.

Appendix: rock records 51–75

Row	Split	v.n obs/pred	v.t obs/pred	omega obs/pred
69	train	2.028 / 1.933	3.880 / 4.060	32.10 / 30.48
70	train	2.883 / 2.503	4.313 / 4.732	37.60 / 35.53
71	train	2.365 / 2.650	3.879 / 4.566	28.70 / 34.29
72	train	2.295 / 2.421	4.609 / 4.751	31.50 / 35.67
73	test	2.337 / 2.818	5.436 / 5.180	29.40 / 38.90
74	test	3.239 / 2.704	5.645 / 5.054	29.70 / 37.95
75	test	2.337 / 2.818	6.109 / 5.180	22.70 / 38.90
76	train	2.097 / 1.244	4.176 / 4.411	39.40 / 33.12
77	train	2.517 / 1.543	4.035 / 4.233	32.60 / 31.79
78	train	1.900 / 1.424	3.927 / 4.377	31.50 / 32.86
79	train	1.909 / 1.525	5.111 / 5.213	31.70 / 39.14
80	train	2.208 / 1.624	4.100 / 5.325	38.40 / 39.99
81	train	2.958 / 1.575	4.362 / 5.292	42.50 / 39.73
82	test	2.767 / 1.875	3.963 / 6.156	40.20 / 46.22
83	test	1.944 / 2.041	5.291 / 6.072	31.30 / 45.59
84	test	3.036 / 2.045	5.309 / 5.702	45.30 / 42.81
85	train	1.239 / 0.699	3.645 / 4.529	40.50 / 34.01
86	train	0.893 / 0.561	4.109 / 4.810	37.60 / 34.47
87	train	0.743 / 0.516	4.885 / 4.541	25.20 / 31.71
88	train	1.423 / 0.618	5.650 / 5.323	32.80 / 37.95
89	train	1.439 / 0.895	4.719 / 5.425	31.60 / 40.73
90	train	1.709 / 0.647	4.014 / 5.956	42.00 / 39.77
91	test	1.993 / 0.962	4.991 / 6.141	43.90 / 46.11
92	test	1.617 / 0.742	6.100 / 5.735	35.60 / 43.07
93	test	1.357 / 0.468	4.224 / 6.865	45.00 / 28.77

Velocities are m/s; angular speed is rad/s. Prediction is the historical sphere proxy, not actual rock geometry. “test denotes the original release-height holdout.

Appendix: all 17 digitized rolling measurements

Specimen	Split	rpm	Measured μ_r	Constant	Fixed curve
moderately_old	train	71.483	0.010602	0.016438	0.011023
moderately_old	held_out	82.770	0.015113	0.016438	0.015763
moderately_old	train	85.771	0.016700	0.016438	0.017024
moderately_old	held_out	94.088	0.019708	0.016438	0.020517
moderately_old	train	100.872	0.022013	0.016438	0.023366
old	train	61.694	0.013609	0.017675	0.013740
old	held_out	70.795	0.016700	0.017675	0.016107
old	train	78.984	0.018204	0.017675	0.018236
old	held_out	86.583	0.019708	0.017675	0.020212
old	train	94.094	0.021211	0.017675	0.022164
old	held_out	97.894	0.024302	0.017675	0.023152
new	train	77.077	0.182403	0.231073	0.167046
new	held_out	87.189	0.197341	0.231073	0.225697
new	train	91.376	0.222060	0.231073	0.249979
new	held_out	96.272	0.258371	0.231073	0.278380
new	train	97.975	0.288755	0.231073	0.288257
new	held_out	100.282	0.320945	0.231073	0.301633